Abhed Srivastava

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Education

Birla Institute of Technology, Mesra *2023 -- 2027*
B.Tech, Mechanical Engineering | CGPA: 8.0/10

Maharshi Patanjali Vidya Mandir, Prayagraj *2022*
Class XII, CBSE (PCM)

Projects

Enterprise Data Warehouse and Analytics Pipeline *Nov 2025 -- Dec 2025*
SQL Server, ETL Pipelines, Power BI, Data Modeling, T-SQL

- Architected and implemented a data warehouse using 3-tier Medallion Architecture (Bronze-Silver-Gold) with automated ETL pipelines integrating 2 disparate ERP and CRM source systems.
- Authored T-SQL transformation scripts for data cleansing, normalization, and loading into an optimized Star Schema, improving analytical query performance by 40%.
- Created SQL analytical aggregations and interactive Power BI dashboards tracking customer behavior, product performance, and sales trends across 15+ key metrics to generate actionable business insights and support decision-making.

Customer Churn Analytics: Statistical Drivers & Predictive Modeling *Nov 2025*
Python, Pandas, NumPy, SciPy, Scikit-learn, XGBoost, SHAP, Streamlit, Matplotlib

- Analyzed 7,000+ customer records across 9+ variables to identify churn drivers, applying structured hypothesis testing to derive statistically defensible business insights.
- Validated churn differences using chi-square tests, Welch's t-test, and ANOVA with Tukey HSD, finding significantly higher churn among month-to-month customers (42.7% vs 2.8% for two-year contracts) with non-overlapping 95% confidence intervals.
- Built and benchmarked churn prediction models (Logistic Regression, Random Forest, XGBoost) achieving 0.837 ROC-AUC; used SHAP for feature importance that corroborated statistically validated drivers and supported data-driven recommendations.

Business AI Agent: Multi-Agent Reasoning and Analysis System *Mar 2026 -- Present*
Python, LangChain, LangGraph, LLMs, RAG, FAISS, SQLite, Streamlit

- Developed a multi-agent system with an LLM supervisor that routes natural-language questions to specialized SQL, RAG, or web-search agents via structured output, enabling end-to-end technology insights generation.
- Engineered a custom SQL agent with automatic retries via error feedback and a hallucination-resistant RAG pipeline over internal documents using FAISS vector search.
- Implemented persistent conversational memory via LangGraph checkpointing for multi-turn queries and a Streamlit interface; performed unit testing and iterative refinement.

Inventory Optimization Under Demand Uncertainty *June 2026 -- Present*
Python, Linear Programming, NumPy, Monte Carlo Simulation, Matplotlib

- Formulated a piecewise-linear LP model to optimize order quantities across 6 SKUs under $6,000 budget and 350 sq. ft. storage constraints, maximizing expected profit under demand uncertainty via concave profit functions.
- Stress-tested the plan via 5,000 Monte Carlo simulations, achieving a 9.9% increase in average weekly profit and a 15.9-point improvement in service level over a mean-demand baseline.
- Visualized profitability and service-level trade-offs to support data-driven inventory decision-making and process improvement.

Technical Skills

Languages:	Python, SQL, C, C++, JavaScript (ES6+), Bash
Data & Analytics:	Power BI, Pandas, NumPy, SciPy, SQL Server, MySQL, SQLite, Data Modeling, ETL
ML & AI:	Scikit-learn, XGBoost, SHAP, LLMs, Multi-Agent Systems, LangChain, LangGraph, RAG, FAISS
Optimization:	Linear Programming, Monte Carlo Simulation
Developer Tools:	Git/GitHub, Docker, Linux, FastAPI, Streamlit
Coursework:	Data Structures & Algorithms (DSA), DBMS, Object-Oriented Programming (OOP)

Achievements

Winner, B-Plan Competition *BIT Mesra, 2024*
Conceptualized and pitched an external GPU enclosure for laptops addressing weight, battery life, and cost constraints; won cash prize of INR 10,000.

Positions of Responsibility

Core Team Member, Events Management *Bitotsav 2025*
Served as core team member for BIT Mesra's flagship cultural fest, contributing to organization and on-ground management of 20+ events across multiple venues and coordinating a team of 10+ volunteers.